Predicting Institution-Level Student Debt Burden from Pre-Outcome Institutional Characteristics

SIADS 696: Milestone II — Final Project Report (Team 15)
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Code and data: https://github.com/Raymay3/College-Debt-Burden-Prediction

1. Introduction

1.1 Problem and Motivation

College affordability is usually discussed in terms of how much students borrow, but debt alone does not describe burden. A $25,000 balance is manageable for a graduate earning $70,000 and crushing for one earning $25,000. Federal student loans are sized by cost of attendance, not by what students are likely to earn afterward, so the institutions where the same debt turns into the heaviest burden are not obvious in advance. We therefore define an institution-level **debt burden** that relates the typical debt of an institution's completers to its former students' long-run earnings:

$$Debt\ Burden_i = ln\left(\frac{GRAD_DEBT_MDN_i}{MD_EARN_WNE_P10_i}\right)$$

Here GRAD_DEBT_MDN is the median federal loan debt of students who completed, and MD_EARN_WNE_P10 is the median earnings of former students who are working and not enrolled ten years after entry. The logarithm makes proportional differences additive; the median institution has a ratio of about 0.34 ($\log \approx -1.09$).

Anticipating this burden from information available *before* the outcome is observed would help students, counselors, and policymakers flag institutions whose students are likely to struggle, years before outcomes appear in federal data. We were also motivated by a methodological question: does the conventional split into public, private nonprofit, and for-profit capture the structure that matters?

1.2 Research Questions

RQ1 (supervised). Can institution-level student debt burden be predicted from institutional characteristics observed before student financial outcomes occur, and which characteristics contribute most?

RQ2 (unsupervised). Do U.S. postsecondary institutions form meaningful empirical groups that differ from conventional sector classifications?

RQ3 (integration). Do representations learned through unsupervised analysis add predictive information beyond the original supervised features?

1.3 Approach

We built one institution-level dataset from the College Scorecard, aggregated to the six-digit OPEID (OPEID6) at which debt and earnings are reported, with 2014–15 predictors and every outcome-related variable excluded by a documented leakage list. Part A compares baselines, regularized linear models, Random Forest, and LightGBM under nested cross-validation; Part B applies PCA and compares k-means, Gaussian mixtures, and Ward clustering without showing them the outcome. We then test whether cluster and PCA features improve prediction when refit inside every fold, and score the frozen model once on 660 held-out institutions (workflow in Appendix C, Figure 9).

1.4 Main Findings

Debt burden is substantially predictable. The selected LightGBM model reached a held-out RMSE of **0.281**, MAE of **0.192**, and R^2 of **0.583** (95% bootstrap interval 0.516–0.642) on 660 unseen institutions, against a cross-validated RMSE of 0.365 for the best non-learned baseline (sector × degree means). Federal-loan participation, degree level, tuition, Pell share, and sector were the strongest predictors.

Accuracy is uneven. Errors are 70–120% larger for for-profit, certificate-focused, and very small institutions than for public, bachelor's-granting, and large institutions.

Institutions form four stable types — certificate schools, four-year teaching colleges, doctoral universities, and community colleges (k-means, bootstrap ARI 0.97). They agree only moderately with sector (ARI 0.38): public and private nonprofit four-year institutions mix freely and split by scope rather than control. The four types explain twice as much burden variance as sector (η^2 0.24 vs. 0.12).

The types do not improve prediction. Cluster and PCA features changed LightGBM's RMSE by less than 0.001, well inside fold-to-fold noise, and failed our pre-registered adoption rule. The structure the clusters summarize is already available to the supervised model, so the simpler supervised-only model was retained.

2. Related Work

Prediction of loan outcomes from the College Scorecard. Luo, Zhang, and Mohanty (2018) applied statistical and machine-learning models to 2007–2014 Scorecard data to identify factors associated with institution-level loan *repayment* rates. We differ in four ways: we model debt relative to earnings rather than repayment (distorted by the 2020–2023 payment pause); we use the newer release with ten-year earnings; we restrict predictors to pre-outcome characteristics through a 2,739-variable leakage list; and we add an unsupervised typology and test whether it improves prediction.

Borrower outcomes by institution type. Looney and Yannelis (2015) linked federal loan records to tax data and showed that the rise in student loan default was concentrated among borrowers at for-profit and two-year institutions, who had weak labor-market outcomes after leaving school. Cellini and Turner (2019) found similarly weak earnings for for-profit certificate students. These studies explain outcomes after the fact with individual-level administrative data; we ask how well public, institution-level characteristics *predict* burden in advance.

Institution-level outcome measures. Chetty, Friedman, Saez, Turner, and Yagan (2020) used tax records to estimate earnings and mobility for every U.S. college. The Department of Education's gainful-employment and financial-value transparency rules (U.S. Department of Education, 2023) use a closely related program-level debt-to-earnings ratio as an accountability metric. Our target is an institution-level analogue of that ratio; unlike these efforts, we treat it as a prediction target and evaluate how much of it is knowable from inputs.

Typologies of institutions. The Carnegie Classification groups institutions by expert rules on degree level and research activity (McCormick & Zhao, 2005); Zemsky, Shaman, and Perna (2014) clustered IPEDS data into data-driven market segments. We follow the data-driven approach but also compare clusters with sector by ARI, test their stability, relate them to an outcome they never saw, and test whether they add predictive value.

Contribution. Our contribution is not another Scorecard prediction model but the combination of (1) a debt-to-earnings burden target, (2) strictly pre-outcome, leakage-controlled features at the OPEID6 level, (3) a comparison of diverse supervised model families under nested cross-validation, (4) a stable, interpretable unsupervised typology compared directly against sector labels, and (5) a leakage-safe test of whether that learned structure improves prediction. This is a new project; it does not extend or reuse any previous milestone or course project.

3. Data Sources and Feature Engineering

3.1 Data Sources

This project uses publicly available data from the U.S. Department of Education's “College Scorecard”, which provides institution-level information on postsecondary education in the United States. The dataset includes institutional characteristics, enrollment, tuition, financial aid, academic programs, student debt, and post-enrollment earnings. We obtained a fixed bulk-data release containing compressed CSV files rather than using the College Scorecard API. The release includes institution-level records covering academic years 1996–97 through 2025–26, field-of-study records, and institution identifier crosswalks. The raw files were preserved to support reproducibility.

For predictor construction, we used the 2014–15 institution-level file (MERGED2014_15_PP.csv.gz) and the 2014–15/2015–16 field-of-study file (FieldOfStudyData1415_1516_PP.csv.gz). The field-of-study records were aggregated to the OPEID6 level to characterize institutional program and credential offerings. Institutional variables describe characteristics such as sector, location, degree offerings, enrollment, tuition, and student demographics. These datasets were selected because they describe characteristics that may help explain variation in institutional debt burden without directly relying on subsequent financial outcomes.

3.2 Unit of Analysis and Data Integration

The unit of analysis is the institution identified by its six-digit Office of Postsecondary Education identifier (OPEID6). College Scorecard institution-level records are also identified by UNITID, which can distinguish campuses or branches associated with the same OPEID6. However, certain federal debt and earnings measures are reported at the OPEID6 level and can consequently appear repeatedly across associated UNITID records.

To avoid treating repeated outcomes as independent observations, we consolidated records to one observation per OPEID6. We first examined whether debt and earnings values were consistent across records sharing an identifier, then collapsed duplicated outcome records without averaging repeated values. Historical institutional characteristics were also aggregated to the OPEID6 level, and field-of-study information was summarized by institution before being merged with the institutional predictors.

The resulting modeling dataset contains 3,296 institutions, each represented by one OPEID6 identifier, a continuous debt-burden outcome, and 35 approved predictors.

3.3 Outcome Construction and Data Cleaning

We defined institution-level student debt burden as the natural logarithm of the ratio of median debt among completers to median earnings ten years after entry:

$$Debt\ Burden_i = ln\left(\frac{GRAD_DEBT_MDN_i}{MD_EARN_WNE_P10_i}\right)$$

Here, GRAD_DEBT_MDN represents median debt among students who completed their studies, while MD_EARN_WNE_P10 represents median earnings among students working and not enrolled ten years after entry. Dividing debt by earnings provides a measure of financial burden relative to economic outcomes, rather than considering debt amounts in isolation. The logarithmic transformation reduces skewness and expresses proportional differences on a continuous scale suitable for regression.

College Scorecard includes privacy-suppressed and unavailable values, including PS, PrivacySuppressed, NA, and NULL. We converted these entries to missing values and retained observations with valid, strictly positive debt and earnings values for outcome construction. Institutions lacking the information necessary to calculate the target were excluded from the supervised modeling sample.

3.4 Feature Engineering and Leakage Prevention

We constructed predictors from historical institutional characteristics and aggregated field-of-study information. The final 35 predictors represent several broad categories, including institutional structure and location, enrollment and student composition, tuition and financial aid, degree offerings, and academic program diversity. Field-of-study records were summarized into institution-level features describing program availability and credential types, allowing program-level information to be incorporated without treating individual programs as separate institutions.

A central consideration was preventing target leakage. Since the objective was to predict financial outcomes using earlier institutional characteristics, we excluded variables directly describing debt, earnings, repayment, default, and later student outcomes from the predictor matrix. The two variables used to construct debt burden were also excluded. Using historical predictors helps maintain the intended distinction between information available before the outcome and information used to evaluate it.

Missing predictor values were handled using model-specific preprocessing approaches. For linear models, designated skewed numeric predictors were log-transformed using log1p, missing numeric values were imputed using the median, missingness indicators were added, and numeric features were standardized using StandardScaler. Random Forest models used median imputation and missingness indicators without standardization, while LightGBM handled missing values natively without requiring median imputation or feature scaling. Where applicable, learned preprocessing operations were fitted within the training folds to prevent information leakage from validation observations. The held-out test set remained excluded from all preprocessing and model-selection decisions. A complete list of the 35 predictors is provided in Appendix A.

3.5 Training and Test Data

After constructing the final modeling dataset, we established a reproducible 80/20 training and held-out test split using random_state=42. The split was performed at the OPEID6 level so that each institution appeared in only one partition. The resulting dataset contains 2,636 training institutions and 660 held-out test institutions.

The held-out test set was reserved for final evaluation, while model development, preprocessing, and hyperparameter selection were performed using the training data. Learned preprocessing operations were fitted within the training workflow to avoid introducing information from the test set. This design supports a more reliable assessment of predictive performance on institutions not used during model development.

4. Supervised Learning: Methods and Evaluation

4.1 Methods

Workflow: All model development used the training set and the testing set was reserved for final evaluation. We created a shuffled 5-fold outer split so that models can be compared fold by fold. The best model was then re-tuned and refit on the full training set.

Feature representation: Models use the 35 predictors. Each model family received the representation that suits its mechanism. The *linear models* got log1p-transformed features and one-hot encoded categorical codes. *Random Forest* gets median imputation plus indicators, without scaling or log transforms. *LightGBM* uses the raw features and handles missing values natively.

Methods and justification: We compared three model families with different mechanisms:

- **Regularized linear regression (Ridge, Lasso, Elastic Net):** The penalties handle the strongly correlated price and enrollment-share variables, and Lasso tests whether a sparse feature set is enough. These models show us how much of the signal is linear.
- **Random Forest:** It captured non-linearities and interactions without specifying them and reduces variance by averaging.
- **LightGBM:** It uses gradient boosting of shallow trees fitted sequentially to the residuals to reduce bias. It performs strongly on mid-sized tabular data and handles missing values natively.

Hyperparameter tuning: All searches minimized inner-CV fold's RMSE. The linear models used grid. The ensembles used randomized search over larger, continuous spaces. Random Forest (300 trees) used 20 draws and LightGBM used 40 draws over number of trees (200–1,500), learning rate (0.005–0.1), leaves (4–48), minimum leaf size, row and column subsampling, and L2 penalty. The final LightGBM uses 963 trees, a learning rate of 0.0056, 39 leaves, min_child_samples = 29, subsample = 0.76, and colsample_bytree = 0.43.

4.2 Overall Results

Metrics: RMSE is the primary metric because it penalizes large errors. RMSE is a multiplicative error and **MAE** describes typical error with less influence from extreme cases. **R^2** is scale-free to allow comparison across baselines and target definitions. Values are mean ± SD over the five outer folds (Table 1; all models in Appendix C, Table 10).

Table 1. Nested 5-fold CV performance (training set, n = 2,636)

Family	Model	RMSE	MAE	R^2
Baseline	Mean	0.422 ± 0.011	0.339 ± 0.010	0.00 ± 0.00
Baseline	Sector × degree mean	0.365 ± 0.016	0.278 ± 0.011	0.25 ± 0.04
Linear	Lasso (best of Ridge/Lasso/EN; all 0.290)	0.290 ± 0.008	0.216 ± 0.005	0.53 ± 0.03
Bagged trees	Random Forest	0.275 ± 0.011	0.200 ± 0.008	0.57 ± 0.02
Boosted trees	**LightGBM (selected)**	**0.274 ± 0.007**	**0.200 ± 0.006**	**0.58 ± 0.01**

Every learned model beat every baseline in every fold. LightGBM reduced RMSE by 25% relative to the sector × degree lookup and more than doubled the explained variance. The tree ensembles beat the linear models by about 0.015 RMSE and LightGBM beat Lasso in all five folds. Random Forest and LightGBM are statistically similar but we selected LightGBM for its lower mean RMSE.

4.3 In-Depth Evaluation of LightGBM

Feature importance and ablation: Permutation importance on held-out folds and mean |SHAP| (Figure 1) rank the same top six features in the same order: PCTFLOAN, HIGHDEG, TUITIONFEE_IN, UGDS_ASIAN, PCTPELL, and PREDDEG. Shuffling federal-loan participation raises RMSE by 0.035. This non-linearity explains the gap between the tree and the linear models. Twelve features add less than 0.001 RMSE. Removing PCTFLOAN, the predictor closest to the outcome costs only 0.004 RMSE (Table 2).

Table 2. Ablation and sensitivity (5-fold CV, LightGBM hyperparameters fixed)

Variation	RMSE	ΔRMSE	R^2
All 35 features	0.272	—	0.583
− ADM_RATE, SAT_AVG	0.274	+0.002	0.578
− CONTROL	0.275	+0.002	0.575
− PCTFLOAN	0.276	+0.004	0.571
Target: all-borrower debt ÷ 10-yr earnings	0.257	n/a	0.573
Target: completer debt ÷ 6-yr earnings	0.280	n/a	0.592

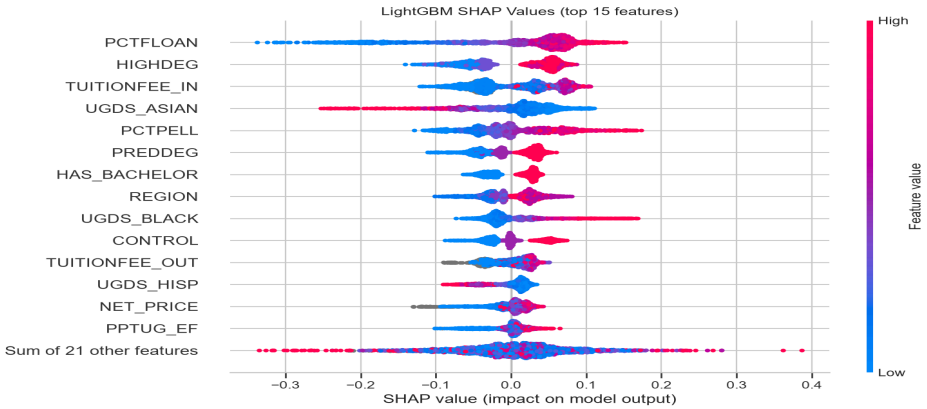


Figure 1. LightGBM SHAP beeswarm (top 15 features).

Sensitivity analysis:

- The per-fold searches chose quite different configurations and yet outer-fold RMSE stayed within 0.265–0.281
- Changes are at most 0.004 RMSE (Table 2).
- LightGBM remains the best model (R^2 0.57–0.59) and beats Lasso by 0.06–0.07 R^2.

Tradeoffs:

- Lasso is fully transparent at R^2 0.53 while LightGBM gains about 5 points of R^2 but needs SHAP to explain it.
- Learning curves (Figure 2) show Lasso plateaued by about 1,000 institutions and LightGBM was still improving at 2,100 so more data would help it.
- The overall R^2 of 0.58 masks large variations in sub-groups. R^2 is 0.72 for bachelor's institutions but only 0.24 for for-profits and 0.20 for certificate institutions.

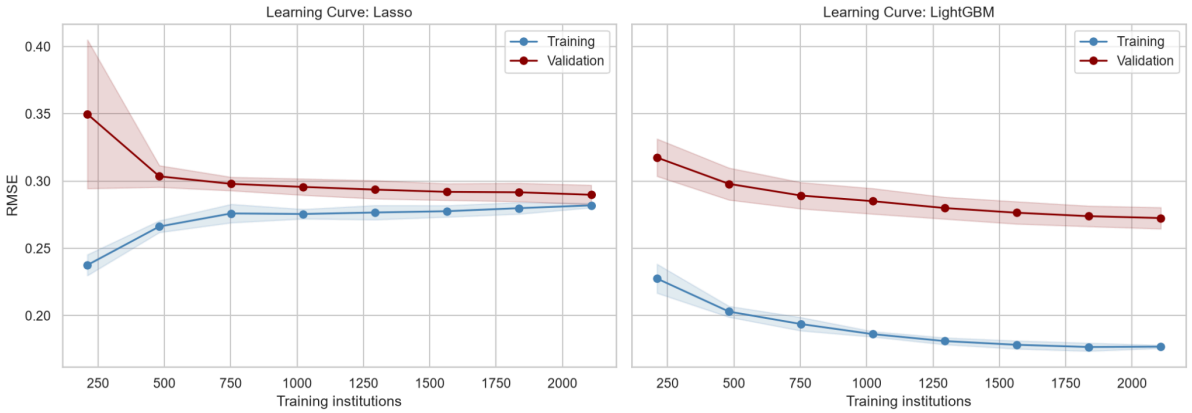


Figure 2. Learning curves for Lasso and LightGBM.

4.4 Failure Analysis

Table 3 shows four representative out-of-fold failures. Field-of-study and cohort-size fields are used for diagnosis only and they are not model inputs.

Table 3. Representative failures (debt-to-earnings ratio, actual vs. predicted)

Institution	Profile	Actual	Pred.
Helene Fuld College of Nursing (NY)	Nonprofit with 1 program and 69% of students borrow	0.14	0.50
Unitek College (CA)	For-profit certificate, health programs	0.13	0.40
Laurus College (CA)	For-profit, 88% students borrow and no tuition data	1.05	0.30
FL School of Traditional Midwifery	Only 49 students and all price data missing	0.98	0.30

Observations:

1. Helene Fuld has a profile the model associates with high burden but its graduates are registered nurses earning about $111k. The model knows the school offers one program, not which one. Health-focused institutions have an RMSE of 0.35 vs. 0.26 for the rest.
2. Unitek's for-profit label pushes its prediction up even though its outcomes resemble a public program.
3. Laurus completers earn $31k against $32k of debt. Since the results are compressed toward the for-profit average, we have a wrong prediction.
4. The midwifery school's median debt comes from only 22 completers and its price predictors are all missing.

Future improvements (not implemented): We can add field-of-study from the Scorecard Field-of-Study files or occupational wage data, add regional labor-market wages, weight institutions by cohort size and fit subgroup-specific models for for-profit and certificate institutions.

4.5 Summary

- LightGBM reaches a nested-CV RMSE of **0.274 ± 0.007** (R^2 **0.577**) versus 0.365 (R^2 0.25) for a sector × degree lookup table.
- Tree ensembles beat linear models in every fold by approx. 0.015 RMSE due to non-linear effects. LightGBM was selected for its lower variance and native missing-value handling.
- Loan participation, degree level, tuition and student composition are main predictors but no single feature or group is essential and results are robust.
- The model is reliable for bachelor's, public and nonprofit institutions but not very strong for for-profit and certificate institutions.

5. Part B — Unsupervised Learning

5.1 Goal

Part B asks whether institutions form meaningful types that cut across sector labels and differ in burden. **Debt burden is never used to form clusters**, and **sector (`CONTROL`) is excluded**, so agreement with sector is a finding, not an input; REGION is excluded as a nominal code. Only the 2,636 training institutions are used.

5.2 Feature Preparation

From the 33 remaining predictors, a training-only filter dropped features with > 40% missing (ADM_RATE, SAT_AVG), near-zero variance (none), or |r| > 0.9 (TUITIONFEE_IN, r = 0.945 with cost; HIGHDEG, r = 0.907 with PREDDEG). Dollar and count features were logged, a missing flag added for out-of-state tuition, values median-imputed, and all 30 features standardized. The steps are scikit-learn transformers in src/unsupervised.py, so they can be refit inside folds (Section 6).

5.3 Dimensionality Reduction: PCA

Institutional characteristics are strongly correlated, which lets redundant dimensions dominate distances; PCA decorrelates and denoises them and gives a compact space for clustering. We kept 13 components (85.8% of variance) (Figure 3a): the proposal allowed 85–90%, the scree curve has no elbow after PC2, and the three extra components for 90% add only ~1.5% each. The components are interpretable (Figure 3b): PC1 (27%) is degree level and program breadth; PC2 (14%) contrasts high price and loan participation with large, part-time, open-access enrollment; PC3–PC4 (13%) are racial and ethnic composition. Institutions differ first in *what kind of education* they provide and second in *how students pay for it*.

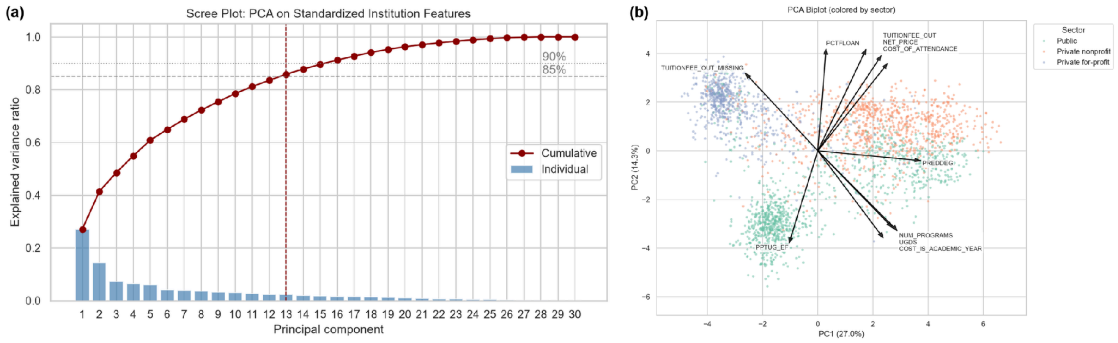


Figure 3. (a) Scree plot: individual and cumulative explained variance; 13 components reach 85%. (b) Biplot of the first two components colored by sector, with the ten strongest feature loadings.

5.4 Clustering Methods

We compared three methods with different assumptions on the 13 PCA scores for k = 3–12:

- **k-means** partitions institutions into compact, roughly spherical groups around centroids. It is fast, deterministic given a seed (20 restarts), and gives a natural assignment rule for new institutions.
- **Gaussian mixture models** (full covariance, 3 restarts) are probabilistic: clusters can be elliptical and differently sized, membership is soft, and BIC provides a likelihood-based guide to k.
- **Ward agglomerative clustering** builds a hierarchy by repeatedly merging the pair of groups that least increases within-cluster variance, with no centroid initialization, and shows how groups nest.

5.5 Cluster Evaluation and Model Selection

We used silhouette (higher is better) and Davies–Bouldin (lower is better) as internal metrics, k-means inertia and GMM BIC as method-specific guides, the smallest cluster size, and bootstrap stability: each method was refit on 30 resamples for k = 3–8, and its assignments compared with the reference solution by adjusted Rand index (ARI). Figure 4 shows the sweep for all three methods; Table 4 compares the best candidates.

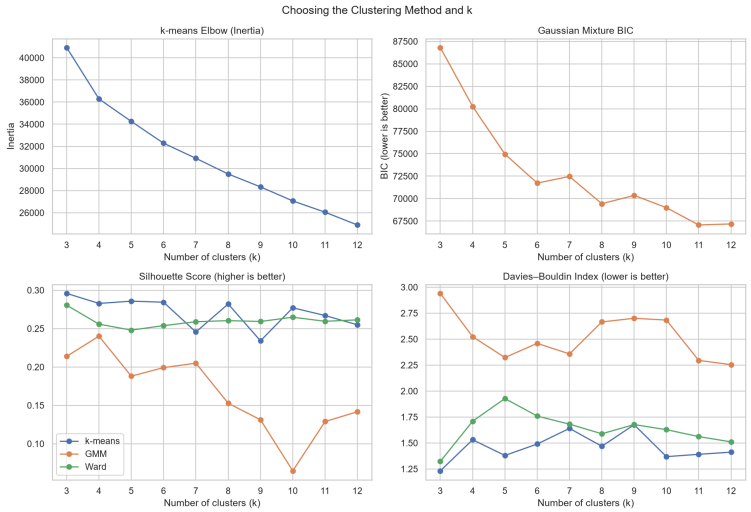


Figure 4. Clustering metrics for k = 3–12: k-means inertia (elbow), GMM BIC, and silhouette and Davies–Bouldin for all three methods.

Table 4. Best candidate solutions by method (training set, 13 PCA components).

Method	k	Silhouette ↑	Davies–Bouldin ↓	Bootstrap ARI ↑	Smallest cluster	ARI vs. sector
k-means	3	0.296	1.232	0.989	687	0.422
k-means (selected)	**4**	**0.283**	**1.531**	**0.965**	**609**	**0.384**
k-means	5	0.286	1.380	0.921	11	0.381
Gaussian mixture	4	0.240	2.523	0.702	444	0.322
Ward	3	0.281	1.322	0.800	614	0.348
Ward	4	0.256	1.709	0.737	577	0.323

k-means had the best silhouette at most k, the lowest Davies–Bouldin at every k, and by far the highest stability (ARI 0.97 at k = 4) versus Ward (0.73–0.80) and GMM (0.53–0.70). GMM's BIC kept falling to k = 11–12, a common artifact of full-covariance mixtures. Silhouettes of 0.25–0.30 indicate moderate structure with soft boundaries. We chose **k-means, k = 4** over the slightly better-scoring k = 3 for interpretability: k = 3 merges all 1,203 four-year institutions, while k = 4 separates doctoral universities from smaller four-year colleges. From k = 5, k-means only splits off an 11-institution cluster of Pacific Islander–serving institutions, too small to be a useful type. Burden was not used to choose k.

5.6 Cluster Profiles

Cluster profiles (Table 5; heatmap in Appendix D, Figure 22) support four plain-language types. Certificate schools are tiny, single-program, and 86% for-profit, with the highest Pell share. Four-year teaching colleges are small bachelor's institutions, 70% nonprofit, with the highest cost and net price. Doctoral universities are large and split between public and nonprofit. Community colleges are 96% public, half part-time, low-cost, and rarely borrow. Racial composition differs by less than 0.3 SD across clusters: the types are defined by scope, price, and enrollment pattern, not demographics.

Table 5. Cluster profiles (training set medians or shares). Burden is shown for description only.

	Certificate schools	Four-year teaching	Doctoral universities	Community colleges
Institutions	685	690	609	652
Public / nonprofit / for-profit	9% / 6% / 86%	20% / 70% / 10%	46% / 51% / 2%	96% / 3% / 1%
Median undergraduates	182	1,440	5,398	3,856

Median cost of attendance	$18,837	$32,324	$27,450	$12,134
Pell share / federal-loan share	61% / 64%	41% / 66%	34% / 56%	41% / 23%
Part-time share	0%	7%	12%	50%
Median programs / offer graduate	1 / 1%	31 / 66%	57 / 100%	38 / 3%
Median debt burden (log)	−1.22	−0.79	−0.97	−1.38

5.7 Comparison with Sector Labels

Agreement with sector is moderate: ARI = 0.38 (NMI 0.41). Two clusters are nearly single-sector, but public and private nonprofit four-year institutions are mixed together and separated by scope rather than control (Figure 5); public institutions appear in all four clusters. The clusters align more with predominant degree (ARI 0.50), matching PCA: they are a *combination* of degree level, scope, price, and sector rather than a relabeling.



Figure 5. UMAP projection of the 13 PCA components (visualization only; clusters were formed in PCA space), colored by k-means cluster (left) and by sector (right). The large four-year region interleaves public and nonprofit institutions and is divided by scope rather than by sector.

5.8 Relationship Between Clusters and Debt Burden

Because burden played no part in forming the clusters, differences in burden are evidence that pre-outcome structure relates to later outcomes (Figure 6; Kruskal–Wallis $p < 10^{-150}$). Clusters explain 23.6% of burden variance, twice sector's 11.7%. Burden is lowest at community colleges and highest at four-year teaching colleges, which combine the highest prices with moderate earnings. Clusters also separate burden *within* sectors: for-profits in the four-year cluster (n = 68) have median burden −0.71 versus −1.19 for for-profit certificate schools. The fair benchmark is degree level: four clusters beat predominant degree (16.7%) and nearly match ten sector × degree cells (25.8%).

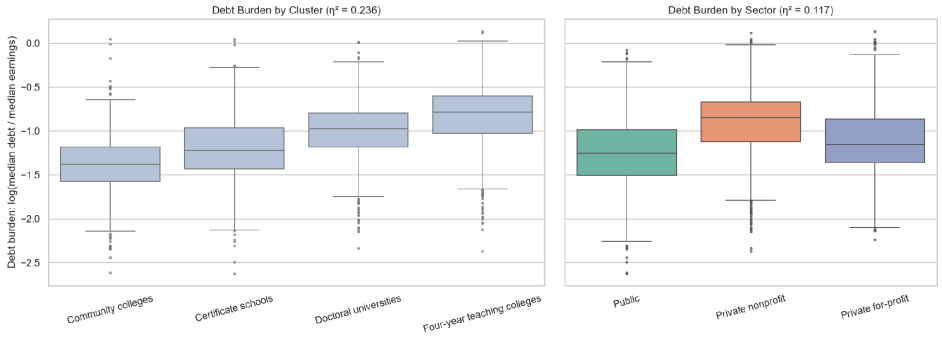


Figure 6. Debt burden by cluster (left, $\eta^2 = 0.236$) and by sector (right, $\eta^2 = 0.117$). Burden was not used to form the clusters.

5.9 Sensitivity Analysis

Bootstrap resampling (Figure 7a) showed stability when the *sample* changes. To test *analysis choices*, we refit k-means (k = 4) with different PCA dimensionality and with one feature group removed, comparing each with the selected clusters by ARI (Figure 7b; Appendix D, Table 17). Retaining 80–95% of variance, or skipping PCA, reproduces the clusters (ARI ≥ 0.97); so does removing demographics (0.97). Removing price (0.84) or aid (0.93) moves some institutions, while removing program mix and degree level reshapes the solution (0.54): degree level is the typology's backbone. In every variant, agreement with sector stays moderate (ARI 0.37–0.41) and clusters explain 21–24% of burden variance. Method and k sensitivity are covered by Table 4 and Figure 4.

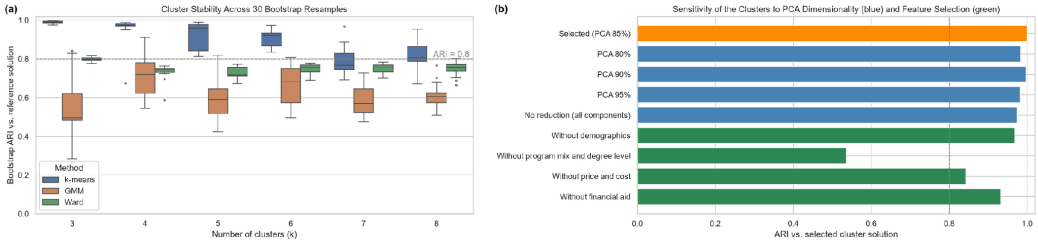


Figure 7. (a) Bootstrap ARI against the reference solution for each method and k (30 resamples). (b) ARI of k-means (k = 4) variants against the selected solution under different PCA dimensionality (blue) and feature removal (green).

6. Integration of Supervised and Unsupervised Learning

Strategy: We tested whether the institution types found by the unsupervised analysis (Part B) add predictive information beyond the original 35 features. The selected Part B pipeline was placed inside the LightGBM pipeline and it was refit on each training fold. Validation institutions were assigned to the nearest training-fold centroid and the debt burden never entered the clustering, thereby ensuring that the comparison is free of leakage.

All variants used the Part A hyperparameters and the same five folds. The pre-defined rule was to adopt cluster features only if the mean RMSE improvement exceeded the SD of the per-fold differences and the integrated model improved in at least 80% of folds. Table 6 summarizes the results.

Table 6. Integration results (5-fold CV, training set, ΔRMSE = paired model vs. supervised only)

Feature representation	RMSE (mean ± SD)	ΔRMSE (mean ± SD)	Folds improved	Rule met?
Cluster-mean baseline (4 groups, no other features)	0.369 ± 0.014	n/a	n/a	n/a
LightGBM, supervised features only	0.2723 ± 0.0098	n/a	n/a	n/a
+ membership + distances (k = 4)	0.2718 ± 0.0089	−0.0005 ± 0.0022	4 / 5	No
+ membership only	0.2721 ± 0.0092	−0.0002 ± 0.0007	3 / 5	No
+ centroid distances only	0.2715 ± 0.0093	−0.0008 ± 0.0023	4 / 5	No
+ PCA scores	0.2718 ± 0.0079	−0.0005 ± 0.0045	3 / 5	No
Repeated CV, 5 × 5 folds (membership + distances)	0.2714 vs. 0.2715	−0.0001 ± 0.0016	11 / 25	No
Lasso + membership + distances	0.2885 vs. 0.2898	−0.0013 ± 0.0020	4 / 5	No

Results:

- The clusters are a good predictor on their own. Predicting each institution's cluster mean out of sample gives an RMSE of 0.369 (R^2 0.24). This beats sector baseline (0.397) and nearly matches the ten sector × degree cells (0.365).
- The clusters add nothing that satisfied our condition to modify the supervised model**.** Every block changed RMSE by less than 0.001 and it was always smaller than the fold-to-fold variation so no variant met the rule.
- Varying k from 2 to 10 gave no improvement in results and hence the number of clusters we create does not have an effect on the resultant model. Hence, we picked k = 4 as the best.

- **Interpretation:** The cluster information is a good predictor but it is redundant. For each model, it added the following value
  - LightGBM showed that the integrated and initial supervised predictions are almost identical (r = 0.993).
  - The trees used the distances as substitutes for the degree-level, price and composition features that the clusters were built from.
  - The Lasso model which does not capture complex interactions had gained the most (approx. 0.0013) but still fell short of our rule and stayed well behind LightGBM.
- We therefore kept the simpler **supervised-only LightGBM** as the final model. The institution types are meaningful but the structure they capture is already learned by the supervised model from the original features.

7. Discussion

7.1 What We Learned from Part A

Characteristics observed years before the outcome explain close to 60% of burden variance, more than double what institution type explains, and the relationship is partly non-linear. **What surprised us** was how little any single feature mattered: removing loan participation — the feature we worried was closest to leakage — cost only 0.004 RMSE, because information is spread redundantly across price, aid, and program variables. We were also surprised that the proposal's payment-based target was the main target shifted by a constant.

Challenges. Many debt measures in the 3,300-column Scorecard do not contain "DEBT" in their names (e.g., monthly payments, portfolio balances), so we classified every variable from the data dictionary instead of searching by name. We also discovered that outcomes repeat across branch campuses and moved the project to OPEID6. **With more time**, we would add program field and institutional-aid features and regional labor-market data, and validate on genuinely later cohorts rather than a random split.

7.2 What We Learned from Part B

Institutions organize first by degree level and program breadth, then by price and how students pay. **What surprised us** was the gap between the clusters' descriptive and predictive value: formed without the outcome, they explain twice sector's share of burden variance yet add nothing to a good supervised model. **Challenges.** The metrics favored k = 3, BIC favored 11–12, and larger k produced tiny clusters, so we added stability and interpretability as explicit criteria; leakage-free integration required packaging the pipeline as reusable transformers. **With more time**, we would cluster on program-level (CIP) offerings, try HDBSCAN, which can leave ambiguous institutions unassigned, and compare our types with the Carnegie Classification.

7.3 Limitations

- Predictions are **institution-level aggregates**, not individual outcomes; median debt and median earnings may not even describe the same students (as at Savannah Technical College; Appendix E, Table 21).
- **Privacy suppression is non-random**: 1,808 institutions, disproportionately small ones, lack an outcome, so the model is least tested where its errors are already largest.
- **Cohort alignment is approximate.** Outcomes pool several entry cohorts; 2014–15 predictors precede the measurement of ten-year earnings but describe institutions while those cohorts were enrolled, not at entry.
- **Missing features**: program field, institutional aid, local labor markets, and macroeconomic context.
- Importance measures are **predictive, not causal**, and test performance on a random split does not establish accuracy for **future cohorts**, especially after the 2025 changes to federal loan limits.

8. Ethical Considerations

Although this project uses publicly available College Scorecard data, ethical concerns remain regarding predictive accuracy, demographic representation, and the potential misuse of institutional classifications. These concerns are particularly important if the models are used to inform educational or financial decisions.

8.1 Supervised Learning Risks

The supervised models predict institution-level student debt burden, but inaccurate predictions could unfairly characterize certain institutions as financially risky. Such classifications could influence prospective students'

perceptions or institutional reputations, particularly if model predictions are presented without uncertainty or context. Furthermore, prediction errors may vary across institution types, potentially producing uneven consequences for public, private nonprofit, and for-profit institutions. To address this, we evaluate predictive performance across institutional sectors and emphasize that predictions should be interpreted alongside their limitations.

Another concern involves demographic predictors, including student racial composition, Pell Grant participation, and first-generation enrollment. Although these characteristics may be associated with institutional debt burden, their predictive relationships should not be interpreted as causal effects or evidence of institutional quality. Importantly, College Scorecard outcomes represent aggregated institutional measures rather than individual student experiences. Consequently, predictions cannot establish the financial circumstances of any particular student. We therefore recommend using these models only for exploratory analysis and decision support, with human review and appropriate disclosure of prediction uncertainty. They should not independently determine financial aid eligibility, accreditation, admissions decisions, or other consequential outcomes.

8.2 Unsupervised Learning Risks

The unsupervised analysis introduces additional ethical considerations because clustering assigns institutions to groups based on selected characteristics. Although these groups may reveal meaningful patterns beyond traditional institutional sectors, cluster membership does not establish institutional quality, educational effectiveness, or student success. Assigning subjective labels such as "high-risk" or "low-quality" could reinforce stereotypes and encourage misleading comparisons between institutions.

Furthermore, cluster assignments depend on methodological choices, including feature selection, scaling, dimensionality reduction, and clustering algorithms. Institutions near cluster boundaries may also have characteristics resembling multiple groups. To reduce these risks, we use descriptive rather than judgmental cluster labels, assess clustering stability, and interpret the resulting groups in terms of their underlying characteristics. Ultimately, these clusters should be treated as exploratory descriptions of institutional similarities rather than definitive rankings or recommendations.

9. Conclusion

This project investigated whether pre-outcome institutional characteristics could predict student debt burden, whether meaningful institutional groupings exist beyond traditional sector classifications, and whether unsupervised learning could improve supervised prediction. Using U.S. Department of Education College Scorecard data, we combined supervised and unsupervised methods to examine institution-level patterns in debt relative to earnings.

Our supervised analysis demonstrated that historical institutional characteristics contain meaningful predictive information about subsequent debt burden. Among the models evaluated, LightGBM achieved the strongest overall performance, with a held-out test RMSE of 0.281 and an R^2 of 0.583. These results indicate that institutional characteristics can explain a substantial portion of the observed variation in debt burden, although prediction errors remain, particularly for certain institution types.

The unsupervised analysis also identified meaningful institutional structure beyond conventional sector labels. Among the clustering approaches evaluated, k-means with four clusters provided a balance of stability, interpretability, and cluster quality. The resulting groups showed only moderate agreement with traditional public, private nonprofit, and for-profit classifications (ARI = 0.38), suggesting that institutional characteristics reveal distinctions not fully captured by sector alone. The clusters also differed in their observed debt burdens, providing additional descriptive insight into institutional variation.

However, incorporating cluster memberships, centroid distances, and PCA-derived features into the supervised models produced only marginal changes in predictive performance. These improvements were insufficient to justify the added complexity, so we retained the supervised-only LightGBM model as our final approach.

Overall, the findings demonstrate the value of combining predictive modeling with exploratory institutional analysis, even when unsupervised representations do not improve prediction. These results support the model as an exploratory tool for studying institution-level patterns in debt burden, not as a prediction of an individual student's financial outcome or a causal assessment of institutional quality.

Statement of Work

All three team members contributed to the project's development, analysis, and final report, with the following division of responsibilities (Table 7).

Table 7. Team member contributions.

Team Member	Contributions
Dominick Devarti	Developed 03_unsupervised_learning.ipynb, including PCA, k-means, Gaussian mixture models, hierarchical clustering, cluster evaluation, stability analysis, and visualizations. Authored the Introduction, Related Work, Part B (Unsupervised Learning), and Discussion sections of the final report.
Gabriella Gass	Developed 01_data_preparation.ipynb, including College Scorecard data cleaning, OPEID6 aggregation, target construction, feature engineering, leakage prevention, exploratory data analysis, and train/test splitting. Independently developed 05_final_evaluation.ipynb, including held-out test evaluation, prediction diagnostics, feature importance, and robustness analyses. Authored the Data Sources and Feature Engineering, Ethical Considerations, Conclusion, and Statement of Work sections of the final report.
Vamsi Krishna Ramesh	Developed 02_supervised_learning.ipynb, including baseline models, regularized regression, Random Forest, LightGBM, cross-validation, hyperparameter tuning, and model comparison. Independently developed 04_integration.ipynb, evaluating the predictive contribution of PCA-derived and clustering-based features. Authored Part A (Supervised Learning) and Integration sections of the final report.

All team members participated in project discussions, coordination, and review of the final deliverables.

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Appendix

Appendix A. Complete Feature Dictionary

All 35 predictors come from the 2014–15 Scorecard institution file (aggregated to OPEID6 as described in Section 3.2) or, for the program-mix group, from the 2014–15/2015–16 field-of-study file. Missing % is for the full modeling sample (n = 3,296).

Table 8. Feature dictionary.

Feature	Description	Aggregation / transform	Missing
CONTROL	Sector: 1 public, 2 private nonprofit, 3 private for-profit	Largest branch	0%
REGION	IPEDS geographic region code (0–9)	Largest branch	0%
PREDDEG	Predominant degree awarded (1 certificate … 4 graduate)	Largest branch	0%
HIGHDEG	Highest degree awarded (0 non-degree … 4 graduate)	Largest branch	0%
NUMBRANCH	Number of branch campuses	Largest branch; log	0%
UGDS	Degree-seeking undergraduate enrollment	Sum; log	0%
UGDS_WHITE, _BLACK, _HISP, _ASIAN, _AIAN, _NHPI, _2MOR, _NRA, _UNKN	Share of undergraduates by race/ethnicity (9 features)	Enrollment-weighted mean	0%
PPTUG_EF	Share of undergraduates enrolled part-time	Enrollment-weighted mean	<0.1%
PCTPELL	Share of undergraduates receiving a Pell grant	Enrollment-weighted mean	0.1%
PCTFLOAN	Share of undergraduates receiving a federal loan (participation, not amount)	Enrollment-weighted mean	0.1%
ADM_RATE	Admission rate	Enrollment-weighted mean	53.0%
SAT_AVG	Average SAT equivalent score of admitted students	Enrollment-weighted mean	64.3%
TUITIONFEE_IN	In-state tuition and fees	Enrollment-weighted mean; log	24.5%
TUITIONFEE_OUT	Out-of-state tuition and fees	Enrollment-weighted mean; log	24.5%
NET_PRICE	Average net price (public NPT4_PUB or private NPT4_PRIV, whichever applies)	Combined; log	3.0%
COST_OF_ATTENDANCE	Cost of attendance (academic-year COSTT4_A, else program-year COSTT4_P)	Combined; log	3.0%
COST_IS_ACADEMIC_YEAR	1 if cost of attendance is academic-year based, 0 if program-year	Derived	3.0%
NUM_PROGRAMS	Number of distinct CIP programs offered	Field of study; log	0%

NUM_CREDENTIAL_LEVELS	Number of credential levels offered	Field of study	0%
HAS_CERTIFICATE, HAS_ASSOCIATE, HAS_BACHELOR, HAS_POSTBACC_CERT, HAS_MASTER, HAS_DOCTORAL, HAS_FIRST_PROFESSIONAL, HAS_GRAD_CERT	Indicator: offers any program at that credential level (8 features)	Field of study (max over rows)	0%

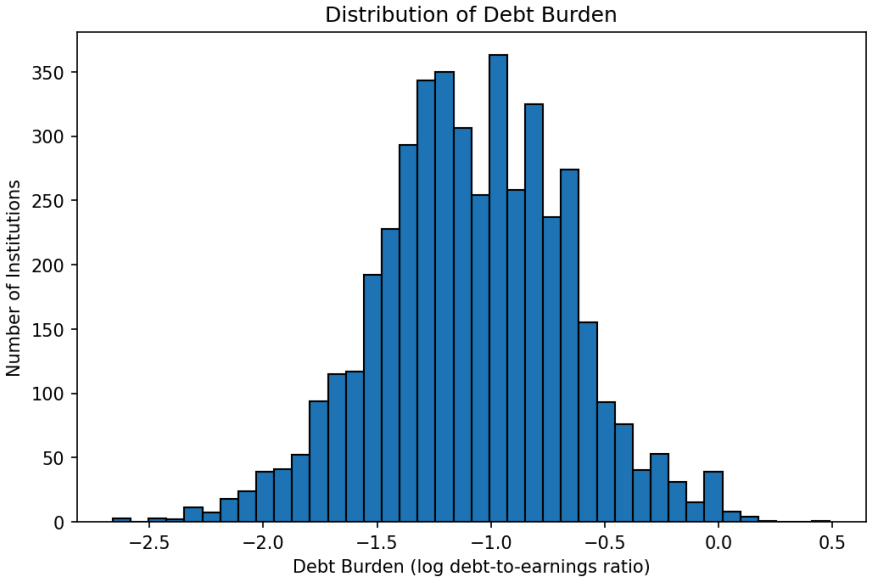


Figure 8. Distribution of the debt-burden target, log(median completer debt ÷ median 10-year earnings), for the 4,465 UNITID records with a valid outcome (before OPEID6 aggregation).

Appendix B. Hyperparameter Search Spaces and Final Values

Every learned model was tuned by an inner 5-fold search scored by RMSE inside each outer training fold (nested cross-validation); the final values below were re-tuned on all 2,636 training institutions.

Table 9. Supervised search spaces (inner 5-fold CV, scored by RMSE) and final values tuned on the full training set.

Model	Search	Space	Final value
Ridge	Grid (26)	$\alpha \in$ logspace(−2, 3)	$\alpha = 25.1$
Lasso	Grid (22)	$\alpha \in$ logspace(−4, −0.5)	$\alpha = 0.001$
Elastic Net	Grid (85)	$\alpha \in$ logspace(−4, 0) (17) × l1_ratio ∈ {0.1, 0.3, 0.5, 0.7, 0.9}	$\alpha = 0.001$, l1_ratio = 0.9
Random Forest	Random (20)	300 trees; max_features ∈ {0.2, 0.33, 0.5, 0.75, 1.0}; min_samples_leaf ∈ {1, 2, 4, 8, 16}; max_depth ∈ {None, 10, 20}	max_features = 0.33; min_samples_leaf = 2; max_depth = 20
LightGBM	Random (40)	n_estimators 200–1,500; learning_rate 0.005–0.1 (log); num_leaves 4–48; min_child_samples 10–80; subsample 0.5–1.0; colsample_bytree 0.3–1.0; reg_lambda 10^{-3}–30 (log)	963 trees; learning_rate 0.0056; 39 leaves; min_child_samples 29; subsample 0.756; colsample_bytree 0.431; reg_lambda 0.0012

Unsupervised settings: PCA retaining 85% of variance (13 components); k-means with 20 initializations; Gaussian mixture with full covariance and 3 initializations; Ward linkage; k = 3–12 for the metric sweep and k = 3–8 for bootstrap stability (30 resamples); random seed 42 throughout.

Appendix C. Supplemental Supervised Results

C.1 Workflow and Cross-Validation (Training Set)

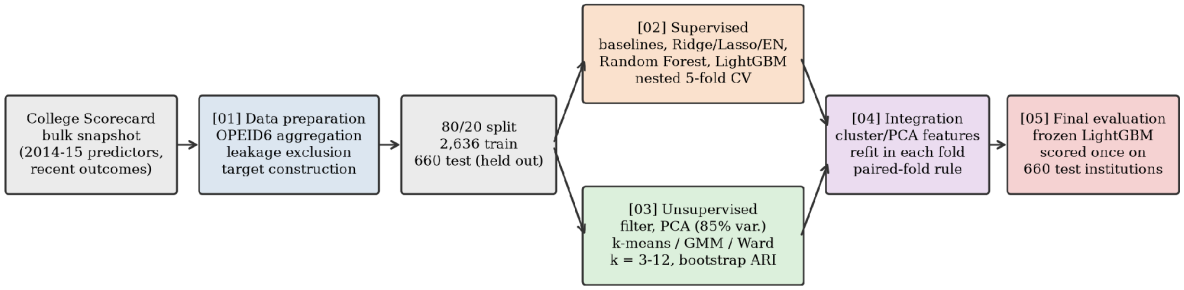


Figure 9. Project workflow. Notebook numbers refer to the repository. Preprocessing, PCA, and clustering are always fit on training data (or training folds) only.

Table 10. Nested 5-fold cross-validation for every model (training set, n = 2,636; mean ± SD across outer folds).

Model	Family	RMSE	MAE	R^2
Mean baseline	Baseline	0.422 ± 0.011	0.339 ± 0.010	-0.001 ± 0.001
Sector baseline	Baseline	0.397 ± 0.014	0.310 ± 0.012	0.116 ± 0.021
Sector × degree baseline	Baseline	0.365 ± 0.016	0.278 ± 0.011	0.251 ± 0.035
Ridge	Linear	0.290 ± 0.009	0.216 ± 0.005	0.526 ± 0.030
Lasso	Linear	0.290 ± 0.008	0.216 ± 0.005	0.527 ± 0.027
Elastic Net	Linear	0.290 ± 0.008	0.216 ± 0.005	0.526 ± 0.028
Random Forest	Bagged trees	0.275 ± 0.010	0.200 ± 0.007	0.573 ± 0.016
LightGBM (selected)	Boosted trees	0.274 ± 0.007	0.200 ± 0.006	0.577 ± 0.013

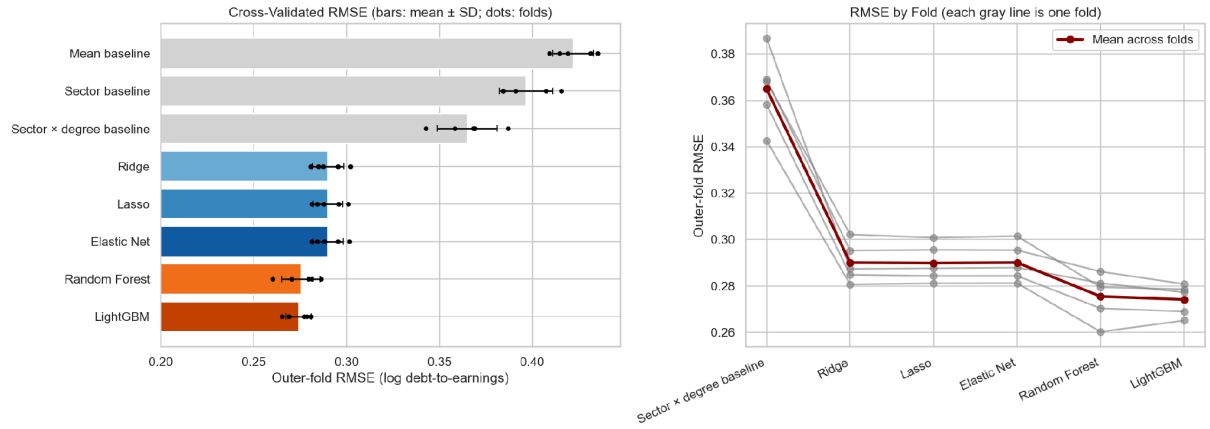


Figure 10. Nested-CV RMSE by model. Left: mean ± SD with individual outer folds as dots. Right: each gray line is one outer fold; every learned model improves on the baselines in every fold.

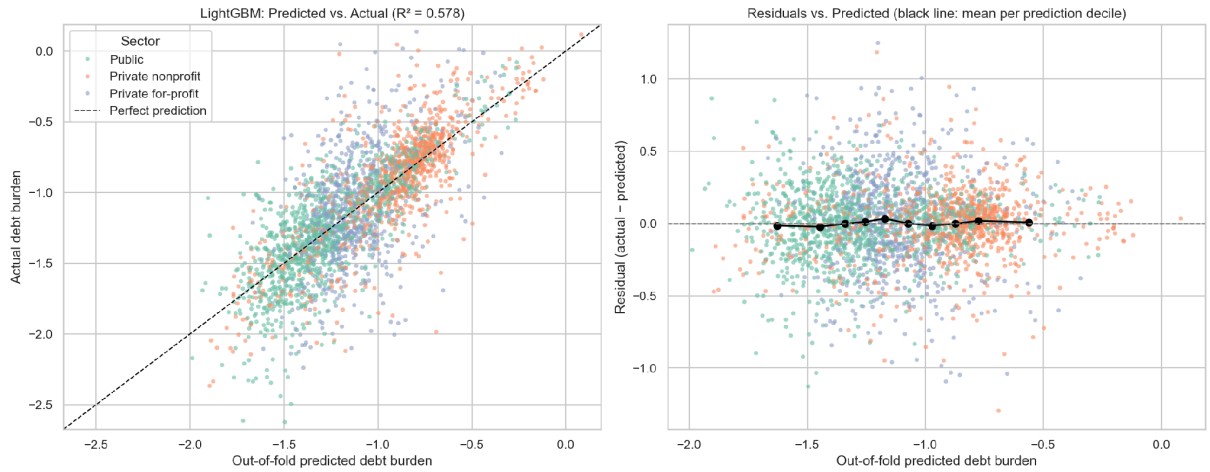


Figure 11. LightGBM out-of-fold predictions on the training set, colored by sector. Left: predicted vs. actual. Right: residuals vs. predictions; the black line is the mean residual per prediction decile.

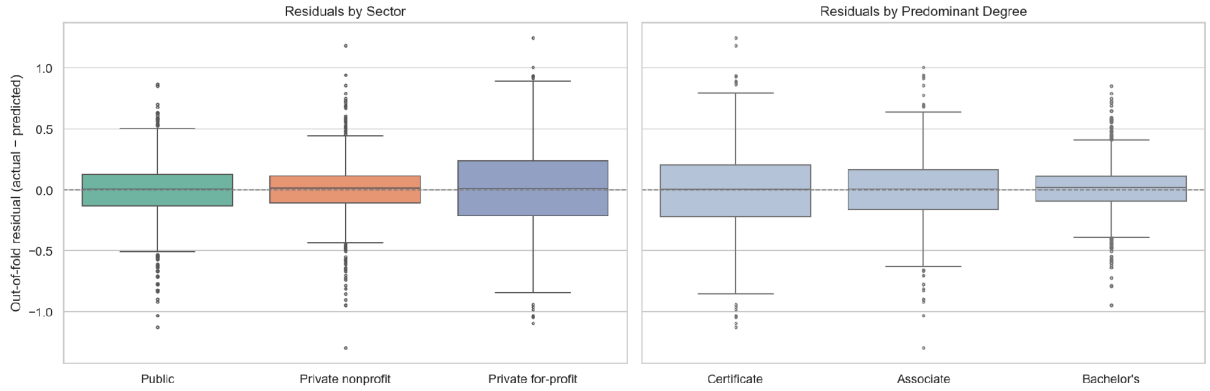


Figure 12. Out-of-fold LightGBM residuals by sector and predominant degree (training set).

Table 11. Out-of-fold LightGBM errors by institution type (training set). Bias = mean residual (actual − predicted).

Grouping	Group	n	RMSE	MAE	Bias	Within-group R^2	Target SD
Sector	Public	1,106	0.236	0.174	-0.008	0.617	0.381
Sector	Private nonprofit	852	0.244	0.171	+0.003	0.641	0.408
Sector	Private for-profit	678	0.356	0.279	+0.015	0.236	0.407
Predominant degree	Certificate	840	0.344	0.266	-0.008	0.200	0.385
Predominant degree	Associate	583	0.292	0.219	-0.004	0.460	0.397
Predominant degree	Bachelor's	1,212	0.200	0.144	+0.011	0.722	0.380

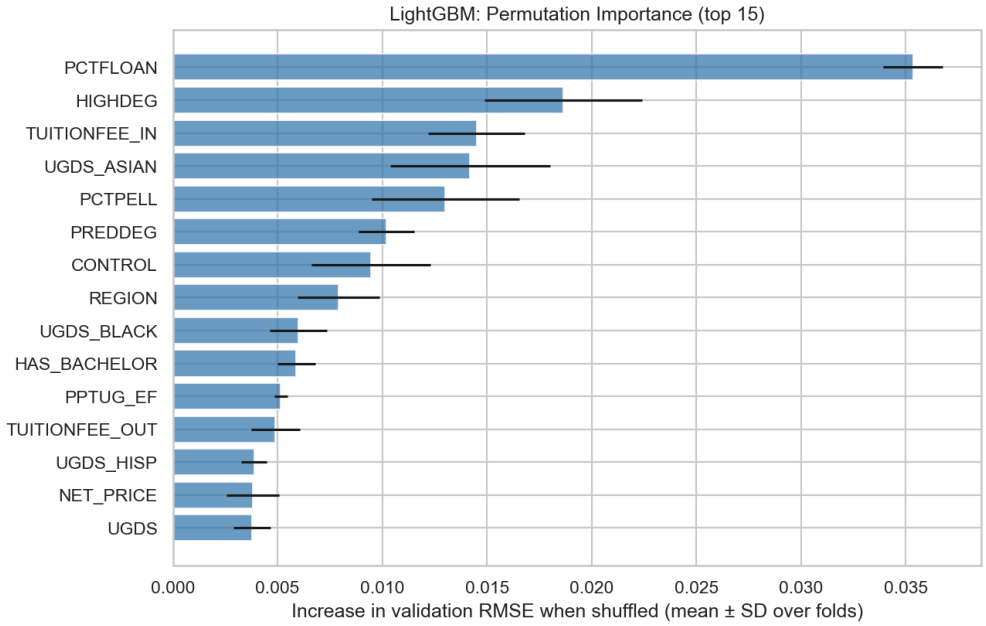


Figure 13. LightGBM permutation importance on the validation part of each outer fold (increase in RMSE when a feature is shuffled; mean ± SD over folds, top 15).

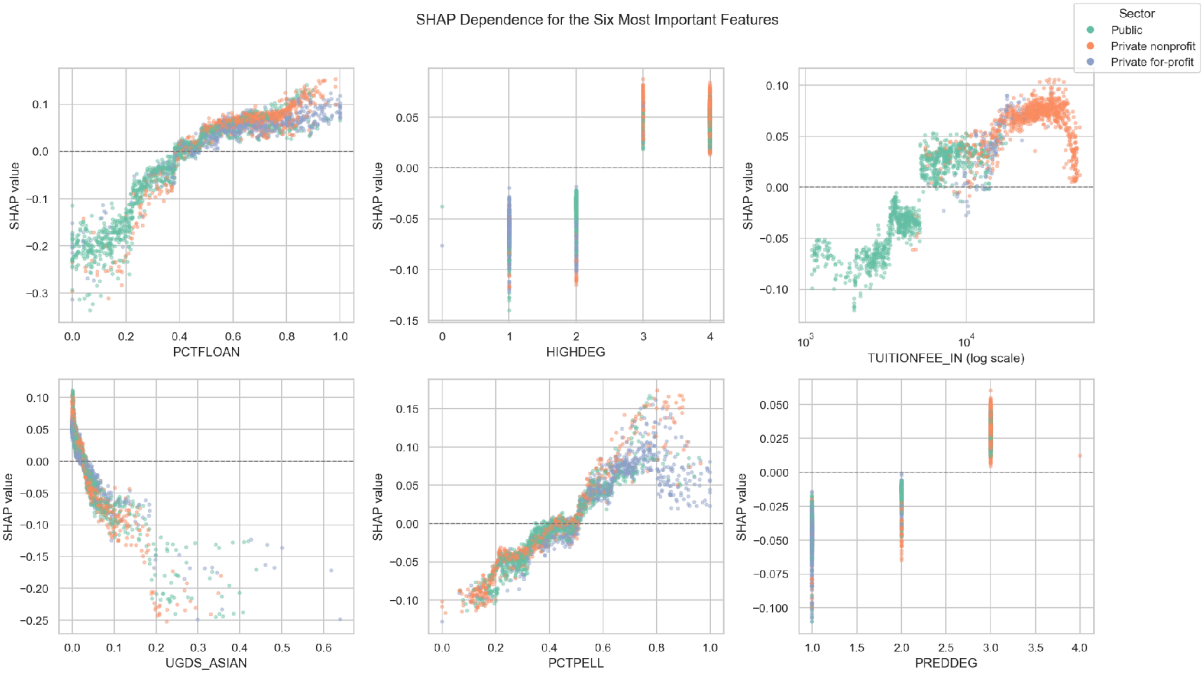


Figure 14. SHAP dependence for the six most important features, colored by sector. Note the non-monotone tuition effect.

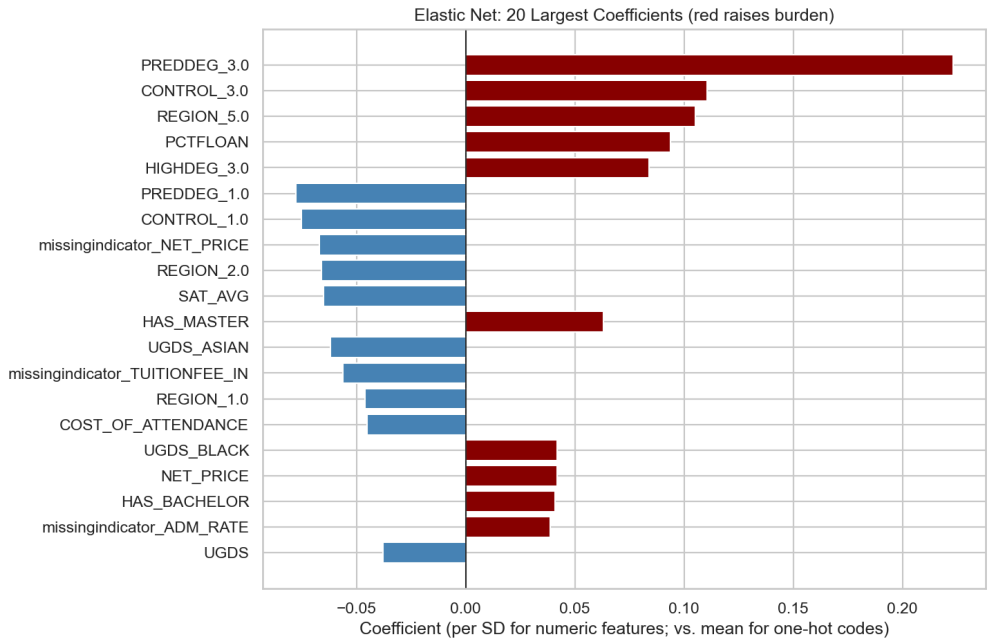


Figure 15. Elastic Net: 20 largest standardized coefficients (red raises predicted burden), for comparison with the non-linear model.

Table 12. Cross-validated R^2 under alternative burden definitions (same folds, hyperparameters fixed at the values tuned for the main target).

Target definition	n	Corr. with main	Sector × degree	Lasso	LightGBM
Completer debt ÷ 10-yr earnings (main)	2,636	1.000	0.251	0.527	0.583
Annual payment ÷ 10-yr earnings (proposal)	2,636	1.000	0.251	0.527	0.583
All-borrower debt ÷ 10-yr earnings	2,636	0.867	0.370	0.513	0.573
Completer debt ÷ 6-yr earnings	2,624	0.962	0.282	0.522	0.592

C.2 Held-Out Test Set (n = 660)

Table 13. Held-out test performance of the frozen LightGBM model with 95% bootstrap intervals (1,000 resamples of the test institutions, model fixed).

Metric	Test estimate	95% interval
RMSE	0.281	0.253–0.307
MAE	0.192	0.176–0.206
R^2	0.583	0.516–0.642

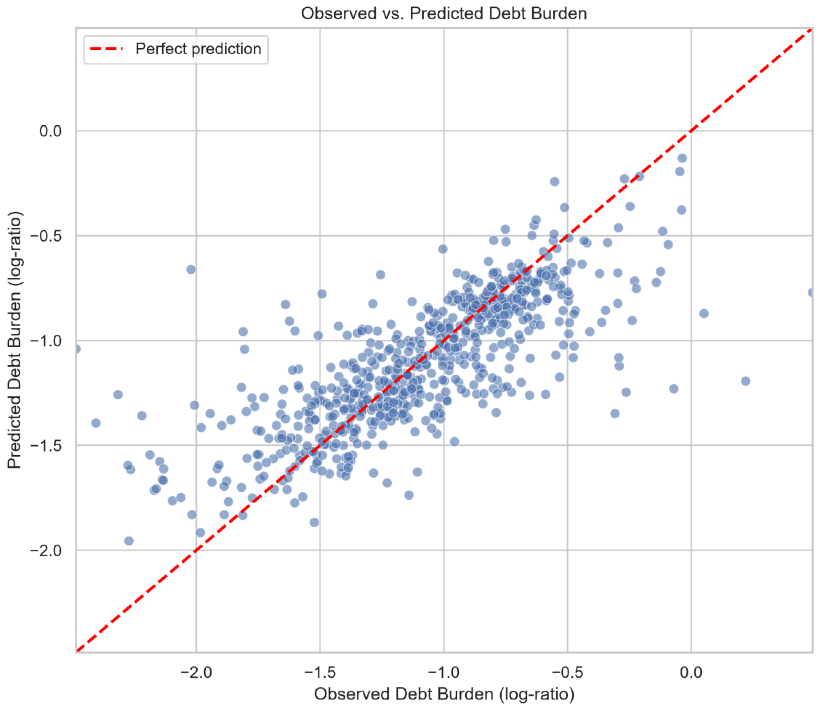


Figure 16. Observed vs. predicted debt burden on the held-out test set (dashed line = perfect prediction).

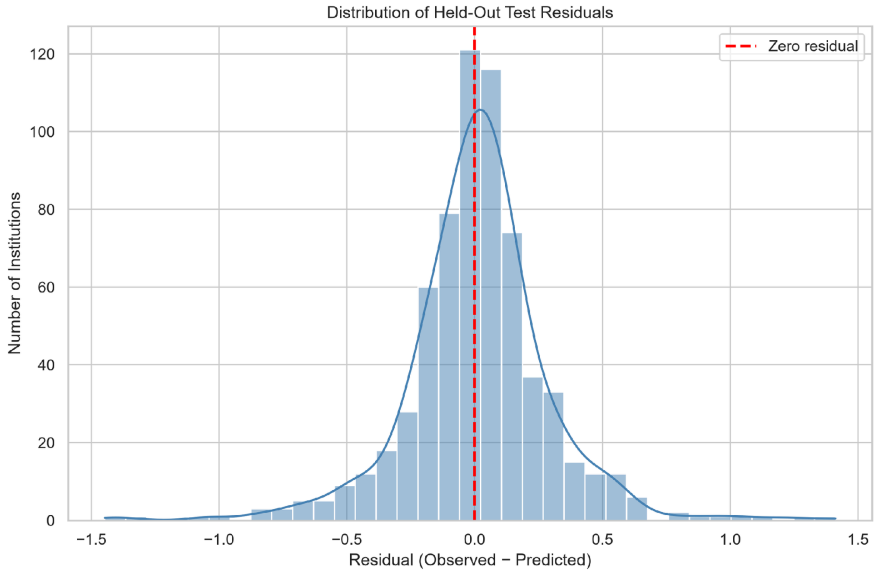


Figure 17. Distribution of held-out test residuals (observed − predicted).

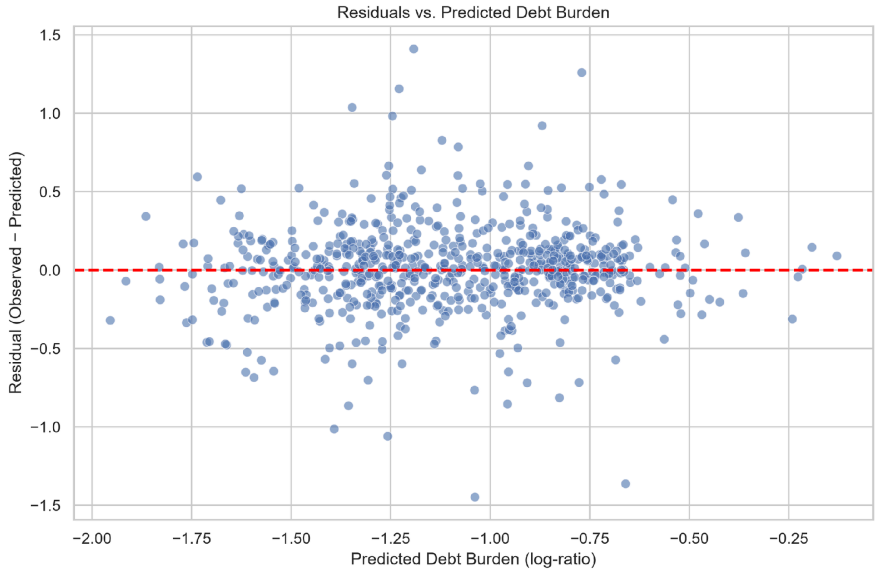


Figure 18. Held-out test residuals vs. predicted debt burden.

Table 14. Held-out test errors by subgroup. Mean residual > 0 means under-prediction on average.

Grouping	Group	n	RMSE	MAE	Mean residual
Sector	Public	263	0.217	0.157	-0.006
Sector	Private nonprofit	235	0.269	0.165	+0.009
Sector	Private for-profit	162	0.375	0.288	+0.047
Predominant degree	Certificate	203	0.363	0.270	-0.021
Predominant degree	Associate	146	0.270	0.191	+0.058
Predominant degree	Bachelor's	311	0.217	0.142	+0.013
Enrollment quartile	Q1: Smallest	165	0.411	0.308	+0.022
Enrollment quartile	Q2: Small	165	0.266	0.181	+0.003
Enrollment quartile	Q3: Large	165	0.203	0.144	+0.017
Enrollment quartile	Q4: Largest	165	0.185	0.137	+0.008

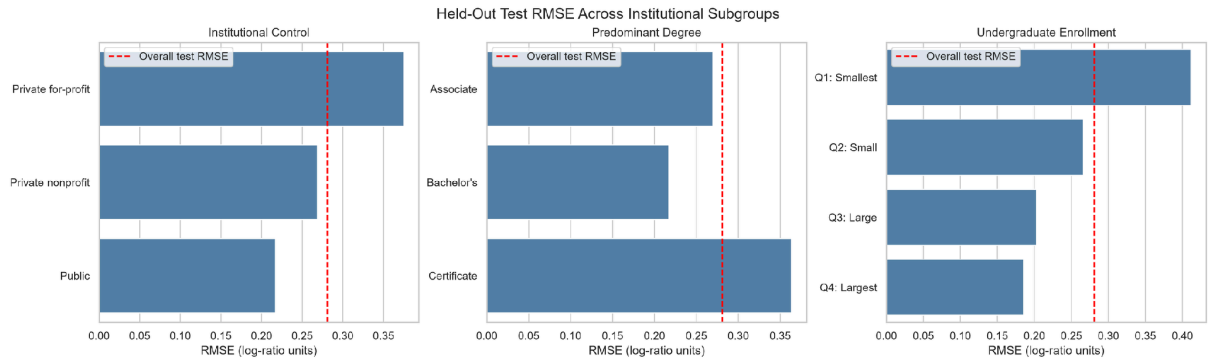


Figure 19. Held-out test RMSE by sector, predominant degree, and enrollment quartile. The dashed line is the overall test RMSE (0.281).

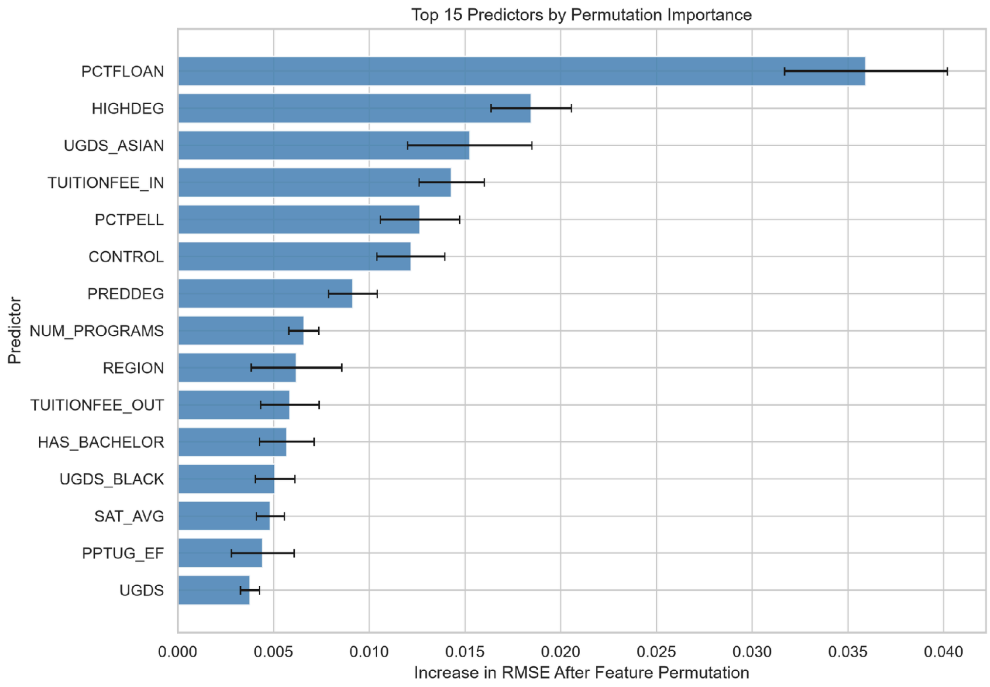


Figure 20. Permutation importance on the held-out test set (increase in RMSE when each feature is shuffled; 10 repeats). The ranking matches the cross-validated importance.

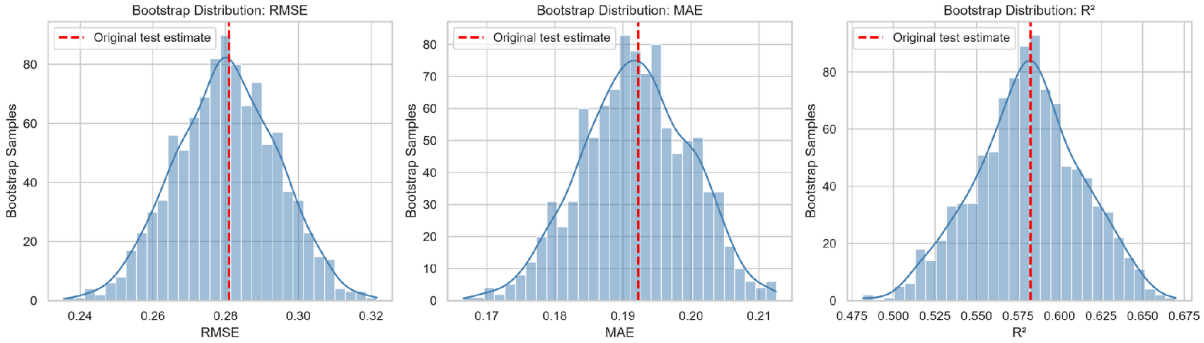


Figure 21. Bootstrap distributions (1,000 resamples of the 660 test institutions) of RMSE, MAE, and R^2; the vertical line is the point estimate.

Appendix D. Supplemental Unsupervised and Integration Results

D.1 Clustering

Table 15. Clustering metric sweep on the 13 PCA components (training set). Silhouette: higher is better; Davies–Bouldin: lower is better.

k	Silhouette k-means	Silhouette GMM	Silhouette Ward	DB k-means	DB GMM	DB Ward	Smallest cluster k-means
3	0.296	0.214	0.281	1.232	2.940	1.322	687
4	0.283	0.240	0.256	1.531	2.523	1.709	609
5	0.286	0.188	0.248	1.380	2.323	1.928	11
6	0.284	0.199	0.254	1.490	2.459	1.759	11
7	0.246	0.205	0.259	1.641	2.358	1.681	11
8	0.282	0.153	0.260	1.468	2.666	1.589	10
9	0.234	0.131	0.259	1.676	2.701	1.677	10
10	0.277	0.065	0.265	1.369	2.683	1.628	3
11	0.267	0.129	0.260	1.391	2.295	1.562	3
12	0.255	0.142	0.261	1.412	2.254	1.511	3

Table 16. Bootstrap stability: mean ± SD adjusted Rand index between each resample's solution and the reference solution (30 resamples).

k	k-means	Gaussian mixture	Ward
3	0.989 ± 0.006	0.534 ± 0.136	0.800 ± 0.010
4	0.965 ± 0.055	0.702 ± 0.095	0.737 ± 0.032
5	0.921 ± 0.067	0.602 ± 0.103	0.731 ± 0.028
6	0.909 ± 0.043	0.662 ± 0.104	0.751 ± 0.024
7	0.786 ± 0.066	0.589 ± 0.075	0.753 ± 0.025
8	0.816 ± 0.063	0.605 ± 0.054	0.752 ± 0.031

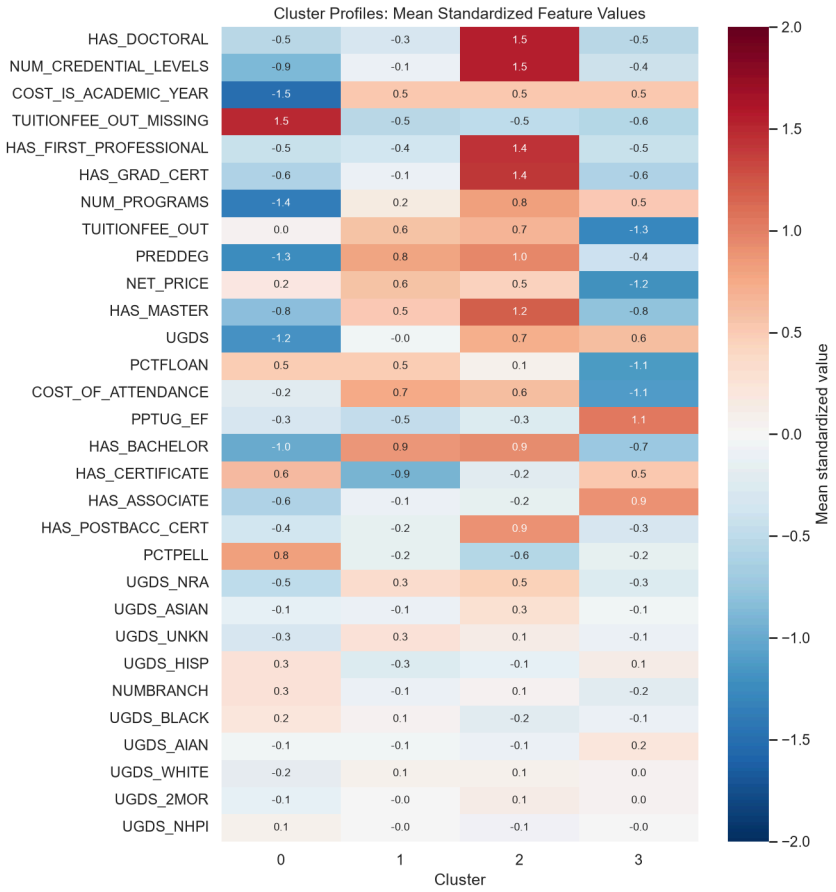


Figure 22. Mean standardized value of each clustering feature by cluster: 0 = certificate schools, 1 = four-year teaching colleges, 2 = doctoral universities, 3 = community colleges.

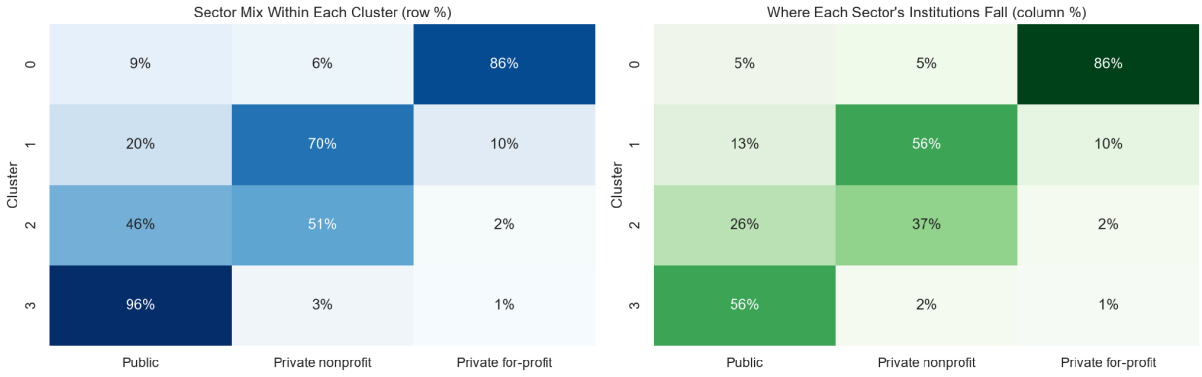


Figure 23. Sector mix within each cluster (row %, left) and distribution of each sector across clusters (column %, right). Clusters numbered as in Figure 22.

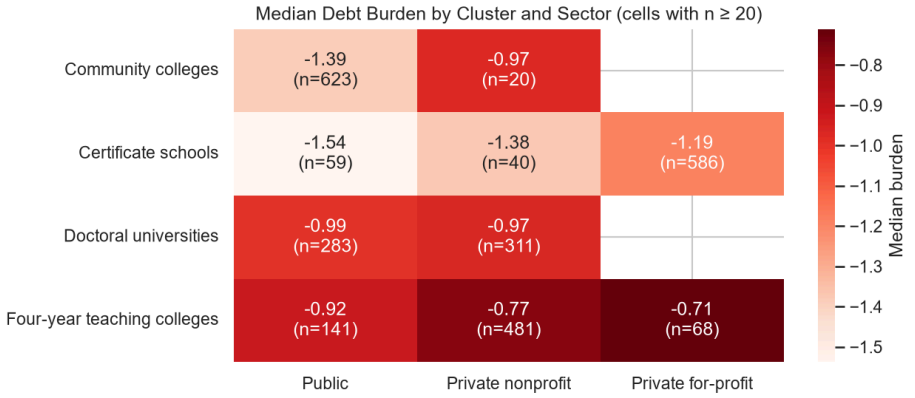


Figure 24. Median debt burden by cluster and sector (cells with n ≥ 20).

Table 17. Sensitivity of the k-means (k = 4) solution to PCA dimensionality and feature selection.

Variant	Dims	ARI vs. selected	ARI vs. sector	Burden η^2	Smallest cluster
Selected (PCA 85%)	13	1.000	0.384	0.236	609
PCA 80%	11	0.983	0.383	0.240	621
PCA 90%	16	0.997	0.385	0.235	608
PCA 95%	19	0.982	0.383	0.235	598
No reduction (30 standardized features)	30	0.974	0.381	0.239	616
Without demographics	8	0.968	0.382	0.234	616
Without program mix and degree level	10	0.536	0.405	0.231	231
Without price and cost	12	0.844	0.382	0.223	584
Without financial aid	13	0.932	0.368	0.212	615

Table 18. In-sample share of debt-burden variance explained by different groupings (training set).

Grouping	Groups	R^2
Sector	3	0.117
Predominant degree	4	0.167
Clusters	4	0.236
Sector + degree	10	0.248
Sector × degree	10	0.258
Sector + clusters	12	0.262
Sector × degree + clusters	30	0.283

D.2 Integration

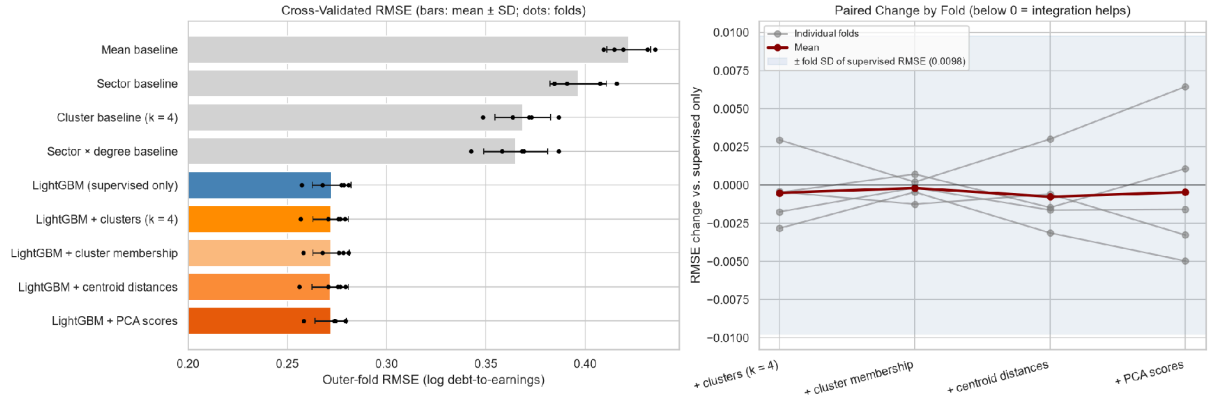


Figure 25. Left: CV RMSE of baselines, the supervised-only model, and integrated variants. Right: paired per-fold RMSE change relative to the supervised-only model; the shaded band is ± the fold SD of the supervised model's own RMSE.

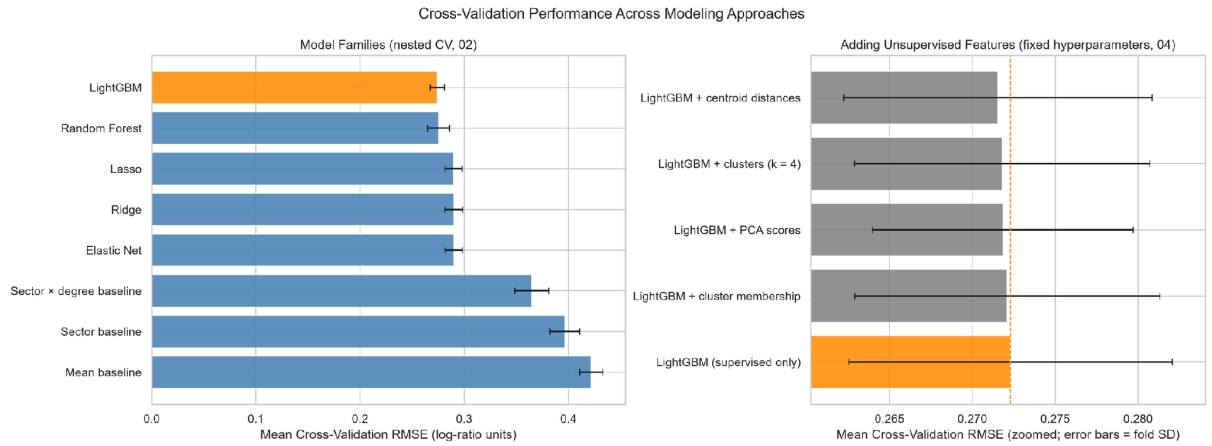


Figure 26. Cross-validation RMSE across modeling approaches: model families under nested CV (left) and supervised-only vs. integrated LightGBM with fixed hyperparameters (right, zoomed).

Table 19. Effect of the number of clusters k on integrated LightGBM (membership + distances; paired change vs. supervised only over the 5 shared folds).

k	CV RMSE	ΔRMSE (mean ± SD)	Folds improved
2	0.2721	-0.0002 ± 0.0021	3 / 5
3	0.2725	+0.0002 ± 0.0024	2 / 5
4	0.2718	-0.0005 ± 0.0022	4 / 5
5	0.2728	+0.0005 ± 0.0019	2 / 5
6	0.2730	+0.0007 ± 0.0025	2 / 5
7	0.2733	+0.0010 ± 0.0019	2 / 5
8	0.2732	+0.0009 ± 0.0017	2 / 5
9	0.2736	+0.0013 ± 0.0032	2 / 5
10	0.2735	+0.0012 ± 0.0025	2 / 5

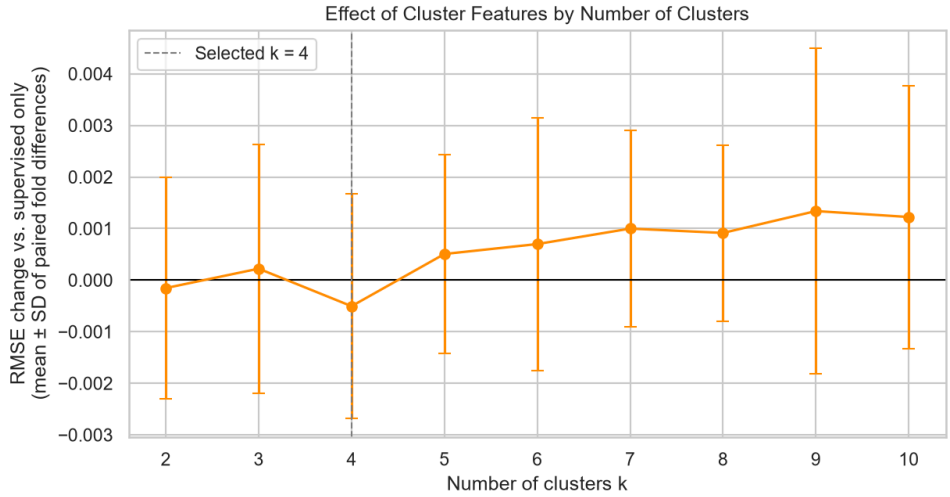


Figure 27. Paired RMSE change from adding cluster features for k = 2–10 (mean ± SD across the five folds).

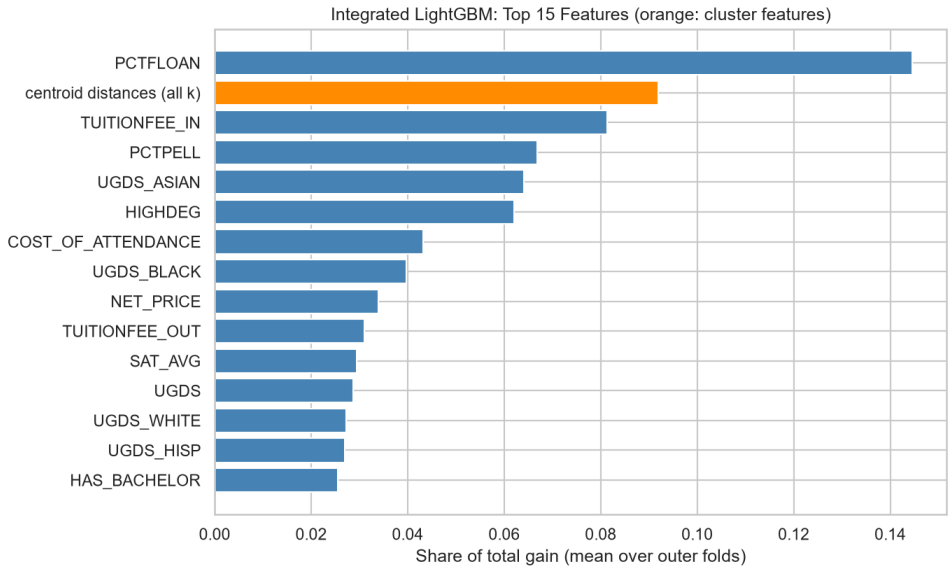


Figure 28. Share of LightGBM split gain in the integrated model (orange: cluster features). About 9% goes to centroid distances and 1% to membership, substituting for the original features.

Table 20. Out-of-fold RMSE by group for the supervised-only and integrated (membership + distances) LightGBM models.

Grouping	Group	n	RMSE supervised	RMSE integrated	Change
Sector	Public	1,106	0.2339	0.2313	-0.0026
Sector	Private nonprofit	852	0.2432	0.2449	+0.0017
Sector	Private for-profit	678	0.3536	0.3533	-0.0003
Predominant degree	Certificate	840	0.3412	0.3400	-0.0011
Predominant degree	Associate	583	0.2904	0.2906	+0.0002
Predominant degree	Bachelor's	1,212	0.1991	0.1986	-0.0005
Cluster	Certificate schools	685	0.3564	0.3558	-0.0006
Cluster	Four-year teaching colleges	690	0.2492	0.2482	-0.0010

Grouping	Group	n	RMSE supervised	RMSE integrated	Change
Cluster	Doctoral universities	609	0.1884	0.1862	-0.0022
Cluster	Community colleges	652	0.2604	0.2615	+0.0011

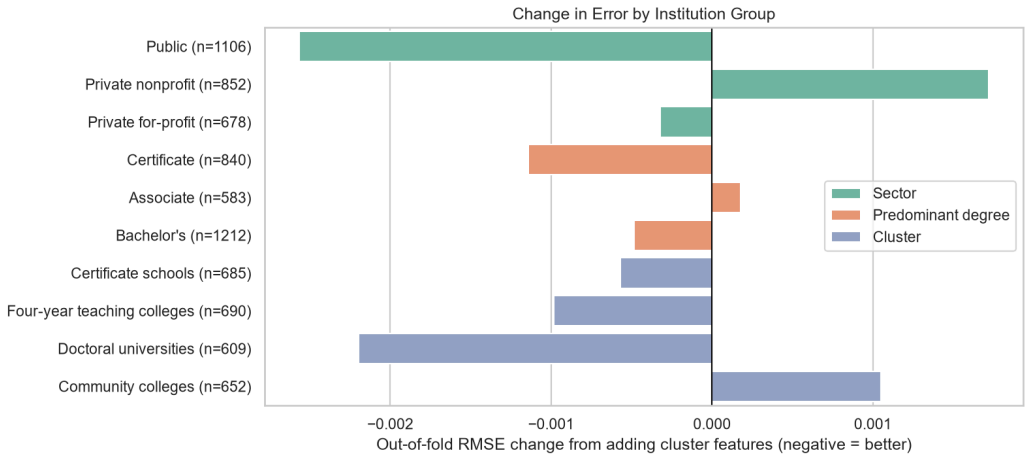


Figure 29. Change in out-of-fold RMSE from adding cluster features, by institution group (negative = better).

Appendix E. Additional Failure Cases

Section 4.4 examines out-of-fold failures on the training set. For completeness, Table 21 lists the fifteen largest errors of the frozen model on the held-out test set, and Figure 30 compares four of them on the ratio scale.

Table 21. Fifteen largest absolute errors on the held-out test set (residual = actual − predicted log burden; > 0 means under-prediction).

Institution	Sector	Degree	UG	PCTFLOAN	Actual	Pred.	Residual
Berea College (KY)	nonprofit	Bachelor's	1,577	0.28	-2.49	-1.04	-1.45
Dorsey School of Beauty (MI)	for-profit	Certificate	172	0.57	0.22	-1.19	+1.41
Albizu University-San Juan (PR)	nonprofit	Bachelor's	370	0.69	-2.02	-0.66	-1.36
Washington Barber College Inc (AR)	for-profit	Certificate	81	0.55	0.49	-0.77	+1.26
Nightingale College (UT)	for-profit	Associate	156	0.91	-0.07	-1.23	+1.16
Savannah Technical College (GA)	Public	Certificate	4,223	0.25	-2.32	-1.26	-1.06
CES College (CA)	nonprofit	Certificate	140	0.41	-0.31	-1.35	+1.04
Hobart Institute of Welding Technology (OH)	nonprofit	Certificate	322	0.31	-2.41	-1.39	-1.01
The Ailey School (NY)	nonprofit	Certificate	96	0.44	-0.26	-1.25	+0.98
Provo College (UT)	for-profit	Associate	364	0.87	0.05	-0.87	+0.92
Meridian Institute of Surgical Assisting (TN)	for-profit	Certificate	383	0.39	-2.22	-1.36	-0.86
Diesel Driving Academy-Shreveport (LA)	for-profit	Certificate	242	0.72	-1.81	-0.96	-0.85
Aviation Institute of Maintenance-Kansas City (MO)	for-profit	Certificate	83	0.55	-0.29	-1.12	+0.83
Albany College of Pharmacy and Health Sciences (NY)	nonprofit	Bachelor's	1,076	0.82	-1.64	-0.83	-0.81
Swedish Institute a College of Health Sciences (NY)	for-profit	Associate	792	0.69	-0.29	-1.08	+0.79

Twelve of the fifteen are certificate or associate institutions and eleven have fewer than 400 undergraduates. The largest over-predictions (Berea College, Albizu University) are institutions whose completers carry unusually little debt despite high sticker prices or high loan participation, a financing pattern the institution-level features cannot see; the largest under-predictions are small, single-program cosmetology and barbering schools whose median earnings come from fewer than 20 former students.

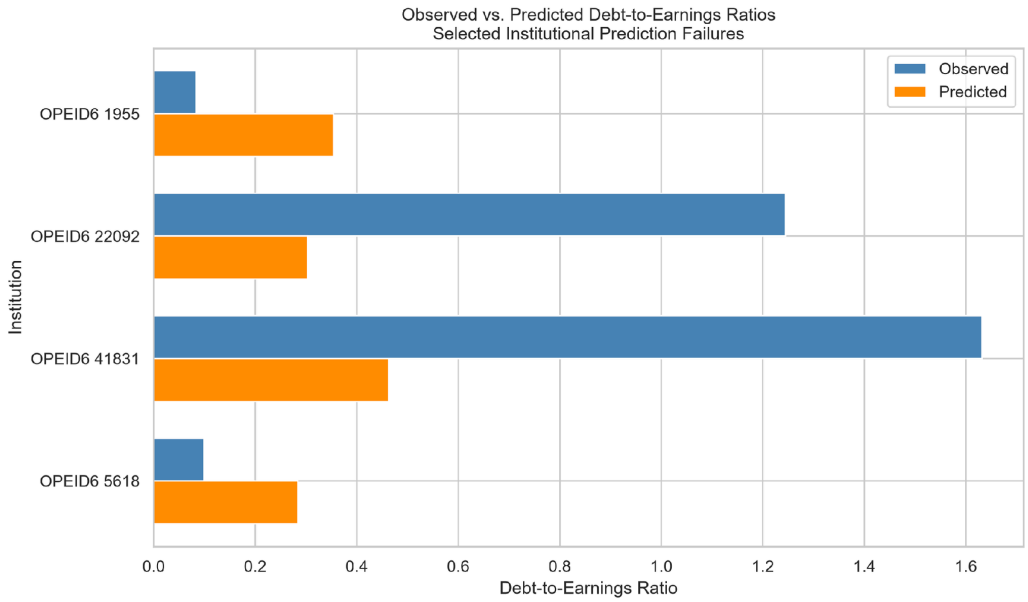


Figure 30. Observed vs. predicted debt-to-earnings ratio for four selected held-out failure cases.